

Li Ye

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EDUCATION

North Carolina State University	08/2025 – 04/2030
Ph.D. in Bioinformatics	Raleigh, NC
University of Michigan, Ann Arbor	08/2021 – 04/2023
M.S. in Bioinformatics (GPA: 4.0)	Ann Arbor, MI
Ocean University of China	08/2016 – 06/2020
B.S. in Biological Science (Rank: 1/55)	Qingdao, China

SKILLS

Python, R, Linux Shell, Slurm, Git, SQL, Docker, C++, MATLAB, LaTeX

PRESENTATIONS

Ye, L. “Machine Learning-Assisted Discovery of Gut Virome Dynamics Across Dietary and Surgical Perturbations” *Poster Presentation, AI in Biology Symposium, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, May 2026.*

PUBLICATIONS

Kodgule R., Goldman JW, Monovich AC, Saari T, Aguilar AR, Hall CN, Rajesh N, Juhi Gupta, Chu S-C A, **Li Ye**, et al. ETV6 Deficiency Unlocks ERG-Dependent Microsatellite Enhancers to Drive Aberrant Gene Activation in B-Lymphoblastic Leukemia. *Blood Cancer Discov.* 2023; BCD-21-0224. doi: 10.1158/2643-3230.BCD-21-0224.

RESEARCH EXPERIENCES

PhD Rotation (Advisor: Dr. Bonnie Hurwitz, Co-advisor: Dr. Alise Ponsero) **03/2026 – 07/2026**
North Carolina State University

Raleigh, NC

- Developed a reproducible shotgun metagenomics workflow on HPC to characterize gut virome dynamics across dietary and bariatric surgery interventions.
- Integrated machine-learning-based bioinformatics tools for viral detection, host and lifestyle prediction, and protein annotation.
- Used Claude Code to convert HPC job scripts from LSF to SLURM.

Research Assistant (Advisor: Dr. Youqiong Ye) **09/2024 – 11/2024**
Shanghai Institute of Immunology & Shanghai Jiao Tong University School of Medicine

Shanghai, China

- Applied a pre-trained ResNet-18 feature extractor and logistic regression classifier to identify and remove artifacts in Visium HD spatial transcriptomics data.

Research Assistant (Advisor: Dr. Marcin Cieslik, Co-advisor: Dr. Russell Ryan) **01/2022 – 05/2022**
Michigan Center for Translational Pathology (MCTP) **Ann Arbor, MI**

- Compared GGAA repeat enhancer activation between specific stages of normal B-cell development and B-progenitor acute lymphoblastic leukemia(B-ALL)
- Isolated different cell populations from published scATAC-Seq datasets of healthy human bone marrow and generated "BigWig" coverage track utilizing tool ArchR.
- Published in *Blood Cancer Discovery*, 2023 (Refer to PUBLICATIONS)

WORK EXPERIENCES

Data Engineer

02/2023 – 05/2025

Aptiv

Troy, MI

- Utilized Python packages (Numpy, SciPy, Matplotlib) to translate vehicle Controller Area Network (CAN) bus data into human-readable values for visualization, encapsulated the code into classes and methods, and converted the script into an executable file (.exe).
- Relied on Suds, a Simple Object Access Protocol (SOAP)-based python web service client, to package Web Services Description Language (WSDL) requests for efficient querying in our web management system (Polarion), and automatically send email notifications with query results upon completion.
- Integrated custom Python scripts into Jenkins, enabling a self-service workflow.
- Developed Python scripts for a data processing pipeline, which analyzes .mat files containing vehicle signal data to evaluate Advanced Driver Assistance System (ADAS) features (Adaptive Cruise Control, Lane Departure Warning, etc.).

Bioinformatics Analyst

05/2022 – 08/2022

Bioinformatics and Sequencing Section at the Michigan Department of Health and Human Services (MDHHS)

Lansing, MI

- Implemented an HIV drug resistance pipeline to identify HIV-1 low frequency drug resistance mutations, optimized the pipeline for speed and generated reports specific to the lab's needs.
- Developed R functions to check whether patient samples contain multiple SARS-CoV-2 infections e.g. Delta and Omicron concurrent infection, which would allow accurate reporting on strains identified in samples, and accurate recommendations for effective treatment.
- Designed a course "Best Practices in Pipeline Development with R" to provide a standard or set of guidelines for R beginners.

Bioinformatics Analyst

04/2021 – 08/2021

Shanghai Biotree Biotech Company

Shanghai, China

- Carried out routine analysis of Non-targeted Metabolomics data by LC-MS and GC-MS. Analyses involved statistical approaches, radar plot, correlation analysis, chord diagram, bubble plot and pathway enrichment analysis.
- Completed subsequent analysis and personalized analysis according to clients' needs, including ROC curve, PERMANOVA and volcano plot with labels.
- Participated in upgrade and development of pipelines utilizing R and R Markdown.

Bioinformatics Analyst

09/2020 – 04/2021

Shanghai OE Biotech Company

Shanghai, China

- Performed analysis of microbial diversity (16S, 18S, ITS) data by next-generation sequencing (NGS). Analyses involved OTU clustering, community's structure, alpha diversity, beta diversity, correlation heatmap, random forest prediction for classification and phylogenetic tree construction.
- Aided in upgrade and development of pipelines utilizing Docker. Utilized Docker to create an image of RDA/CCA analysis and then pushed it to Harbor. Connecting this new Docker image with the Cloud Platform, the company could improve the efficiency and flexibility of subsequent analysis and personalized analysis.

AWARDS

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| • Office of Graduate and Postdoctoral Studies Master's Student Award | 2022 |
| • OUC the First Class Scholarship Award | 2019 |
| • OUC the Second Class Scholarship Award | 2018 |
| • OUC the Third Class Scholarship Award | 2017 |
| • Outstanding Student Award | 2017 |